Yi Hu

+1 825-984-9793 | hu19@ualberta.ca | linkedin | website

Education

PhD in Biomedical Engineer Jan. 2020 – (Expected May 2026)
University of Alberta *Edmonton, Canada*

MSc in Control Theory and Control Engineering Sept. 2017 – Jun. 2020
Sichuan University *Chengdu, China*

B.S. in Process Equipment and Control Engineering Sept. 2013 – Jun. 2027
Hefei University of Technology *Hefei, China*

Personal Summary

Robotics engineer with a PhD in autonomous surgical systems, specializing in trajectory generation, motion optimization, and data-driven learning for robotic ultrasound and endoscopic scanning. Experienced in integrating perception, planning, and policy learning for high-precision autonomous manipulation.

Research Experience

Human-Centered Autonomy Lab Jan. 2024 – Present
Graduate Research Assistant, University of Alberta *Edmonton, Canada*

- Fine-tuned large vision–language–action (VLA) models (Pi0) on real-world robot datasets using NVIDIA A6000 GPUs; deployed on a Franka robot with real-time RGB-D image inference, task success rate about XX%.
- Integrated a 3D spatial representation module into VLA models (**Pi0, OpenVLA**) to enable cross-modal fusion and 3D-aware manipulation; developed a Mixture-of-Experts action head for improved geometric reasoning, achieving XX% and XX% success rates on LIBERO Spatial environments.
- Designed and implemented a cross-embodiment reinforcement-learning framework to transfer human daily-activity demonstrations to robotic surgical tasks; validated on the da Vinci robot for precise tool manipulation and camera-view control.

Telerobotic and Biorobotic Systems Group Jan. 2021 – Dec. 2023
Graduate Research Assistant, University of Alberta *Edmonton, Canada*

- Developed an autonomous ultrasound scanning framework on the Franka robot, integrating **admittance control** and tissue–probe force contact regulation; trained and deployed a **U-Net** model for ultrasound image assessment.
- Implemented a **hand-pose–based teleoperation system** for 6-DoF ultrasound probe manipulation; built a vision pipeline for 3D hand pose estimation and mapped human motion to robot end-effector with safety constraints.
- Implemented a **surgical view control** system with Endoscopic Camera Manipulator on da Vinci robot; applied unsupervised motion primitive learning to optimize camera trajectories in constrained surgical workspaces.
- Developed precise, collision-aware 6-DoF manipulation with Patient Side Manipulator on da Vinci robot; developed smooth trajectory generation and orientation-constrained motion planning algorithms.

Perception, Motion Control and Intelligent Robot Innovation Lab Sept. 2017 – Dec. 2020
Graduate Research Assistant, Sichuan University *Chengdu, China*

- Designed and simulated control algorithms for **fixable-joint manipulators** under actuator saturation and external disturbances, improving trajectory stability and robustness.
- Modeled and controlled a **magnetic-wheel mobile robot**, deriving kinematic and dynamic models for precise trajectory tracking.
- Developed a Unity-based simulation environment for magnetic-wheel robot navigation and visual localization.

Technical Skills

Robots: da Vinci Robot, Franka Robot, Magnetic Mobile Vehicle, PHANToM Premium 1.5A Robot
Languages: English, Chinese, Cantonese